



COMS5023A

Lecturer: Dr Alton Dewa

E-mail: alton.dewa@wits.ac.za

Instructions

1. Use any SQL platform of your choice (e.g., MySQL, SQL Server)
2. Answer all the questions in a Microsoft Word document.
3. For each question:
 - Write the SQL query (code).
 - Copy and paste the SQL code into your Word document.
 - Execute the query in your SQL platform.
 - Take a screenshot of the query output.
 - Insert the screenshot directly below the corresponding SQL code in the Word document.

1. Write a query to display the current date. Your output should be as shown in the screenshot: **(1 mark)**

Results Messages	
	(No column name)
1	2026-08-04 18:20:19.610

2. Write an SQL statement to create a table named Student with the following columns: StudentID, LastName, FirstName, Fees, Gender, Course, DateOfBirth, NSFASPayment. Your output should be as shown in the screenshot: **(5 marks)**

Results Messages							
StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFASPayment

3. Write an SQL INSERT statement to add 15 records to the Student table. Your output should be as shown in the screenshot. **(5 marks)**

Results Messages								
	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1001	Mudau	Rendani	45000.00	Male	Information Technology	2003-05-12	18000.00
2	1002	Netshifhefhe	Lufuno	52000.00	Female	Accounting	2004-08-21	22000.00
3	1003	Mulaudzi	Takalani	61000.00	Male	Engineering	2002-11-03	25000.00
4	1004	Baloyi	Nkateko	38000.00	Male	Business Management	2005-02-18	16000.00
5	1005	Hlungwani	Tsakani	56000.00	Female	Information Technology	2004-07-27	21000.00
6	1006	Mavundla	Mbhazima	47000.00	Male	Education	2003-09-14	19500.00
7	1007	Mashigo	Katlego	34000.00	Male	Marketing	2006-01-30	15000.00
8	1008	Mokgadi	Ntswaki	62500.00	Female	Engineering	2002-06-11	24000.00
9	1009	Mahlangu	Thabo	42000.00	Male	Accounting	2005-10-05	17500.00
10	1010	Mbatha	Babalwa	55000.00	Female	Human Resources	2004-03-22	23000.00
11	1011	Mgangatho	Luyanda	36000.00	Male	Education	2006-05-16	15500.00
12	1012	Ndzuzo	Asanda	64000.00	Female	Information Technology	2002-12-09	24500.00
13	1013	Dlamini	Sibusiso	50000.00	Male	Business Management	2003-04-25	200.00
14	1014	Khumalo	Ayanda	59000.00	Female	Marketing	2005-08-13	22500.00
15	1015	Mthembu	Sfiso	43000.00	Male	Accounting	2004-11-19	17000.00

4. Write an SQL query to display all students whose Fees are between R40,000 and R60,000 (inclusive). Use the BETWEEN operator to retrieve the records from the student table. Your output should be as shown in the screenshot. **(3 marks)**

Results Messages								
	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPaymer
1	1001	Mudau	Rendani	45000.00	Male	Information Technology	2003-05-12	18000.00
2	1002	Netshifhefhe	Lufuno	52000.00	Female	Accounting	2004-08-21	22000.00
3	1005	Hlungwani	Tsakani	56000.00	Female	Information Technology	2004-07-27	21000.00
4	1006	Mavundla	Mbhazima	47000.00	Male	Education	2003-09-14	19500.00
5	1009	Mahlangu	Thabo	42000.00	Male	Accounting	2005-10-05	17500.00
6	1010	Mbatha	Babalwa	55000.00	Female	Human Resources	2004-03-22	23000.00
7	1013	Dlamini	Sibusiso	50000.00	Male	Business Management	2003-04-25	200.00
8	1014	Khumalo	Ayanda	59000.00	Female	Marketing	2005-08-13	22500.00
9	1015	Mthembu	Sfiso	43000.00	Male	Accounting	2004-11-19	17000.00

5. Write an SQL query to display the StudentID, FirstName, LastName, and NSFasPayment of students whose NSFasPayment is between R18,000 and R22,000. Your output should be as shown in the screenshot. **(3 marks)**

Results Messages				
	StudentID	FirstName	LastName	NSFasPayment
1	1001	Rendani	Mudau	18000.00
2	1002	Lufuno	Netshifhefhe	22000.00
3	1005	Tsakani	Hlungwani	21000.00
4	1006	Mbhazima	Mavundla	19500.00

6. Write an SQL query to display all female students enrolled in Information Technology. Your output should be as shown in the screenshot. **(3 marks)**

	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1005	Hlungwani	Tsakani	56000.00	Female	Information Technology	2004-07-27	21000.00
2	1012	Ndzuzo	Asanda	64000.00	Female	Information Technology	2002-12-09	24500.00

7. Write an SQL query to display all students who are studying Accounting or Engineering. Your output should be as shown in the screenshot. **(3 marks)**

	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1002	Netshifhefhe	Lufuno	52000.00	Female	Accounting	2004-08-21	22000.00
2	1003	Mulaudzi	Takalani	61000.00	Male	Engineering	2002-11-03	25000.00
3	1008	Mokgadi	Ntswaki	62500.00	Female	Engineering	2002-06-11	24000.00
4	1009	Mahlangu	Thabo	42000.00	Male	Accounting	2005-10-05	17500.00
5	1015	Mthembu	Sfiso	43000.00	Male	Accounting	2004-11-19	17000.00

8. Write an SQL query to display all students whose LastName starts with the letter 'M'. Your output should be as shown in the screenshot. **(2 marks)**

	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1001	Mudau	Rendani	45000.00	Male	Information Technology	2003-05-12	18000.00
2	1003	Mulaudzi	Takalani	61000.00	Male	Engineering	2002-11-03	25000.00
3	1006	Mavundla	Mbhazima	47000.00	Male	Education	2003-09-14	19500.00
4	1007	Mashigo	Katlego	34000.00	Male	Marketing	2006-01-30	15000.00
5	1008	Mokgadi	Ntswaki	62500.00	Female	Engineering	2002-06-11	24000.00
6	1009	Mahlangu	Thabo	42000.00	Male	Accounting	2005-10-05	17500.00
7	1010	Mbatha	Babalwa	55000.00	Female	Human Resources	2004-03-22	23000.00
8	1011	Mgangatho	Luyanda	36000.00	Male	Education	2006-05-16	15500.00
9	1015	Mthembu	Sfiso	43000.00	Male	Accounting	2004-11-19	17000.00

9. Write an SQL query to display all students whose FirstName ends with 'a'. Your output should be as shown in the screenshot. **(2 marks)**

	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1006	Mavundla	Mbhazima	47000.00	Male	Education	2003-09-14	19500.00
2	1010	Mbatha	Babalwa	55000.00	Female	Human Resources	2004-03-22	23000.00
3	1011	Mgangatho	Luyanda	36000.00	Male	Education	2006-05-16	15500.00
4	1012	Ndzuzo	Asanda	64000.00	Female	Information Technology	2002-12-09	24500.00
5	1014	Khumalo	Ayanda	59000.00	Female	Marketing	2005-08-13	22500.00

10. Write an SQL query to display the FirstName, LastName, and DateOfBirth of students born in the year 2005. Your output should be as shown in the screenshot. **(2 marks)**

	FirstName	LastName	DateOfBirth
1	Nkateko	Baloyi	2005-02-18
2	Thabo	Mahlangu	2005-10-05
3	Ayanda	Khumalo	2005-08-13

11. Write an SQL query to calculate the total amount of Fees payable by all students using the SUM() function. Your output should be as shown in the screenshot. **(2 marks)**

Results Messages	
	TotalFees
1	744500.00

12. Write an SQL query to calculate the total NSFASPayment for all students using the SUM() function. Your output should be as shown in the screenshot. **(2 marks)**

Results Messages	
	TotalNSFASPayment
1	280700.00

13. Write an SQL query to display each Course and the number of students enrolled in that course using the COUNT() function. Your output should be as shown in the screenshot. **(3 marks)**

Results Messages		
	Course	NumberOfStudents
1	Accounting	3
2	Business Management	2
3	Education	2
4	Engineering	2
5	Human Resources	1
6	Information Technology	3
7	Marketing	2

14. Write an SQL query to determine the number of female students in the Student table using the COUNT() function. Your output should be as shown in the screenshot. **(2 marks)**

Results Messages	
	NumberOfFemaleStudents
1	6

15. Write an SQL query to display all students who are not enrolled in Accounting and Engineering. Your output should be as shown in the screenshot **(3 marks)**

Results		Messages						
	StudentID	LastName	FirstName	Fees	Gender	Course	DateOfBirth	NSFasPayment
1	1001	Mudau	Rendani	45000.00	Male	Information Technology	2003-05-12	18000.00
2	1004	Baloyi	Nkateko	38000.00	Male	Business Management	2005-02-18	16000.00
3	1005	Hlungwani	Tsakani	56000.00	Female	Information Technology	2004-07-27	21000.00
4	1006	Mavundla	Mbhazima	47000.00	Male	Education	2003-09-14	19500.00
5	1007	Mashigo	Katlego	34000.00	Male	Marketing	2006-01-30	15000.00
6	1010	Mbatha	Babalwa	55000.00	Female	Human Resources	2004-03-22	23000.00
7	1011	Mgangatho	Luyanda	36000.00	Male	Education	2006-05-16	15500.00
8	1012	Ndzuzo	Asanda	64000.00	Female	Information Technology	2002-12-09	24500.00
9	1013	Dlamini	Sibusiso	50000.00	Male	Business Management	2003-04-25	200.00
10	1014	Khumalo	Ayanda	59000.00	Female	Marketing	2005-08-13	22500.00

Total Marks=40